Summarized Meeting Minutes

Project Name: Financial and Operational Analysis of U.S. Hospitals Using Hospital Provider Cost Report Data (2011–2022)
Course: Capstone Project 2
Group: 8
Institution: St. Clair College
Term: Winter 2026

Date & Time

Consolidated Review (Jan 12 – Mar 31, 2026)

Location

St. Clair College / Microsoft Teams

Meeting Type

In-Class & Online

Attendees

Name	Student ID	Role
Manjunath Muthineni	0878795	Team Leader & ML Lead
Rajesh Thota	0873782	Documentation Lead
Neha	0873038	Research & Domain Lead
Abhishekh Choudhary	0873460	Data Quality Lead
Krishna Chaitanya Venuturumilli	0874277	ML Support & Review Lead
Professor Manjari Maheshwari	—	Faculty Advisor

Agenda Item 1: Project Planning & Methodology (MM1–MM2)

Discussion

- Project scope was clearly defined, and team roles were assigned based on individual strengths to ensure efficient collaboration and accountability
- CMS HCRIS dataset was selected as the primary data source due to its comprehensive financial and operational coverage of U.S. hospitals
- A structured plan was developed to integrate external datasets (BLS, RUCC, Census) in a controlled manner
- The professor emphasized the importance of establishing a strong methodological foundation before model implementation
- The team agreed to focus on evaluation strategy and data understanding prior to developing machine learning models

Decisions Made

- Establish a clear machine learning workflow before implementation
- Prioritize data validation and evaluation strategy over early model development
- Ensure all modeling decisions are well-justified and documented

Action Items

Task	Assigned To
Define ML workflow	Manjunath
Prepare documentation structure	Rajesh
Review domain knowledge	Neha
Validate dataset quality	Abhishekh
Support planning and assumptions	Krishna

Agenda Item 2: Dataset Integration & Feature Engineering (MM2–MM5)

Discussion

- Dataset integration was carried out in a phased approach to minimize risk and ensure data integrity
- Phase 1: Integration of CMS with BLS wage data and RUCC codes
- Phase 2: Integration of Census socioeconomic data after professor approval
- Extensive join validation was performed to avoid duplication, incorrect mappings, and inconsistencies across datasets
- Missing values were analyzed and handled using appropriate strategies such as imputation, exclusion, or use of proxy variables
- RUCC codes were simplified into rural/urban categories to improve interpretability and usability in modeling
- The final dataset structure was refined iteratively and prepared for modeling

Decisions Made

- Adopt a phased dataset integration strategy to ensure reliability
- Retain only validated and meaningful features for modeling
- Ensure strict data quality checks before proceeding to modeling

Action Items

Task	Assigned To
Validate dataset joins	Abhishekh
Document integration process	Rajesh
Review feature relevance	Neha
Finalize dataset structure	Manjunath
Support validation checks	Krishna

Agenda Item 3: Baseline Modeling & Initial Validation (MM3–MM4)

Discussion

- Random Forest baseline models were developed to predict Profit Margin and Charity Care Ratio (CCR), providing an initial benchmark for performance
- Feature importance analysis was conducted to understand the contribution of each variable
- Results indicated that BLS wage variables and Census socioeconomic indicators were among the strongest predictors
- These findings validated the integration of external datasets and confirmed their relevance to hospital financial performance
- The professor reviewed the results and confirmed that the baseline modeling approach was methodologically sound

Decisions Made

- Approve Random Forest as the baseline modeling approach
- Retain external datasets as key predictive features
- Use baseline results as a benchmark for further model improvement

Action Items

Task	Assigned To
Train baseline models	Manjunath
Analyze feature importance	Krishna
Validate domain interpretation	Neha
Document findings	Rajesh
Support testing and validation	Abhishekh

Agenda Item 4: Advanced Modeling & Optimization (MM4–MM10)

Discussion

- Multiple machine learning models were explored, including ensemble methods (Random Forest, ExtraTrees, Gradient Boosting) and linear models (Ridge, Lasso)
- Cross-validation was introduced to ensure reliable performance evaluation and to reduce dependence on a single train-test split
- Sequential Feature Selection was applied to reduce dimensionality while maintaining predictive power
- Hyperparameter tuning using GridSearchCV was performed to optimize model performance
- Ensemble models consistently outperformed linear models, indicating the presence of nonlinear relationships in the data

Decisions Made

- Adopt cross-validated R^2 (CV R^2) as the primary evaluation metric
- Continue focusing on ensemble-based models for improved performance
- Use feature selection and tuning to enhance model generalization

Action Items

Task	Assigned To
Implement advanced models	Manjunath
Perform hyperparameter tuning	Rajesh
Validate model performance	Neha
Review modeling assumptions	Krishna
Support preprocessing and feature selection	Abhishekh

Agenda Item 5: Model Selection & Leakage Removal (MM10)

Discussion

- A comprehensive data leakage audit was conducted to ensure model integrity
- Variables directly derived from target metrics and high-cardinality identifiers were removed to prevent biased predictions
- Model performance was evaluated using cross-validation to ensure robustness and generalizability
- Final comparison confirmed that Random Forest and ExtraTrees provided the best balance of accuracy and stability

Decisions Made

- Final models selected:
    - Random Forest (Profit Margin)
    - ExtraTrees (Charity Care Ratio)
- CV R^2 adopted as the primary model selection metric
- Exclude models that showed higher complexity without performance improvement

Action Items

Task	Assigned To
Finalize model training	Manjunath
Document evaluation results	Rajesh
Validate model assumptions	Neha
Verify feature consistency	Abhishekh
Support model validation	Krishna

Agenda Item 6: Deployment Strategy & Application Development (MM11)

Discussion

- The professor recommended using a simple and user-friendly deployment approach, such as Streamlit or Flask
- The team identified that too many input fields could reduce usability for non-technical users
- A decision was made to simplify the interface by reducing visible inputs and deriving additional features programmatically
- A Streamlit application was developed with prediction functionality, batch processing, and model information sections
- The design focused on clarity, usability, and accessibility for end users

Decisions Made

- Select Streamlit as the primary deployment framework
- Simplify UI using derived features and default values
- Ensure consistency between training and deployment pipelines

Action Items

Task	Assigned To
Develop application	Manjunath
Implement feature derivation	Rajesh
Validate UI usability	Neha
Test application pipeline	Abhishekh
Perform end-to-end testing	Krishna

Agenda Item 7: Final Testing & Submission Preparation (MM11)

Discussion

- The end-to-end pipeline was tested to ensure accurate predictions and system stability
- Column mismatch issues between training and deployment were identified and resolved
- Batch prediction functionality was tested and confirmed to work correctly
- Final documentation and outputs were reviewed to ensure completeness and alignment with academic requirements

Decisions Made

- Confirm system readiness for final submission
- Ensure all deliverables are complete and properly documented

Action Items

Task	Assigned To
Perform final testing	Manjunath
Complete documentation	Rajesh
Validate outputs	Neha
Ensure data consistency	Abhishekh
Verify deployment workflow	Krishna

Additional Notes

- The project followed a structured progression from planning to deployment
- All major decisions were validated through professor feedback
- The final solution balances model performance, interpretability, and usability